Hadi Zaatiti

Research Scientist
Modelling & Simulation of Dynamical Systems

Doctor of engineering with 10 years of combined industry and academic experience, specialized in the modelling and simulation of dynamical systems across the automotive, military aerospace and healthcare industries.

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French citizen, 33 years old



Experience

Research Scientist
Jan 2024 to present, full-time
New York University, Abu Dhabi

Magnetoencephalography (MEG) and electroencephalography (EEG) labs, part of the BioMedical Imaging Core at NYUAD.

- **Experimental design**: Work with research groups to design and implement MEG and EEG experiments, matching the recording setup to the neuroscience question being asked.
- **Systems**: Maintain, calibrate and monitor the MEG, EEG and fMRI-compatible EEG systems, and adapt hardware and software as research needs change.
- **Analysis pipelines**: Build the data-processing pipelines that take raw recordings to clean, reproducible results (MNE-Python, FieldTrip, BIDS).
- **Research projects**: Lead projects from first idea to publication, for peer-reviewed papers and applied clinical contexts.
- **Clinical collaboration**: Epilepsy project with Cleveland Clinic Abu Dhabi (see Projects).

System Designer, MBSE & Simulation Specialist
Feb 2022 to Jan 2024, full-time
Airbus Defence & Space

Model Manager for two programs: FCAS (Future Combat Air System) and ED (EuroDrone). Also served as MBSE & Simulation Tutor at Airbus Academy.

- **MBSE processes**: Design technical processes for writing requirements and modeling operational, functional and technical architecture elements.
- **Systems engineering deployment**: Apply architectural frameworks such as UAF and UBM (Cameo Systems Modeler, Enterprise Architect, Doors, RPE, Jira, Confluence).
- **Mission simulation**: Define and simulate mission scenarios at different levels of abstraction (Cameo Simulation Kit, MATLAB).
- **Automate generation & verification**: requirements/documentation generation and automated testing/ verification of models (JavaScript, NoMagic API, Jenkins, GitHub, Bitbucket).
- **Train and standardize**: deliver MBSE and simulation training to future complex-system designers; contribute to state-of-the-art systems engineering (INCOSE, AFIS), ISO standards, and certification programs (CSEP).

System Architect for Autonomous Vehicles & Data Scientist
Nov 2018 to Jan 2022, full-time
Institute of Research and Technology SystemX

Architect for autonomous transportation systems in collaboration with RATP, Transdev, SNCF, Renault, PSA, Navya and EasyMile.

- **Autonomous shuttles**: Designed, specified and modeled a complete autonomous shuttle system with project partners, ensuring correct behavior via verification and simulation techniques.
- **Representation learning on manifolds (with Stanford and Inria)**: Worked on data representation on manifolds (mainly graphs) as the 3rd contributor to geomstats.ai, and on learning methods for complex system models (simple, timed and hybrid automata).

System Modeling and Simulation Teacher
Nov 2018 to Jan 2022, part-time
École Centrale de Paris, CentraleSupélec

Taught 4th-year engineering students system modeling, requirements engineering and simulation through a practical case: capturing stakeholder needs, building an architecture, evaluating technical solutions and justifying design choices.

Research Engineer
Nov 2015 to Nov 2018, full-time
Commissariat of Atomic Energy (CEA)

Modelling, simulation and formal verification of hybrid dynamical systems.

- Modeled complex systems as finite state machines coupled with differential equations (hybrid automata) and formally verified them via Satisfiability Modulo Theories (SMT and SAT).
- Developed an approach that automates the abstraction of such models to verify safety properties expressed through constraints.
- Implemented the approach in C++ (over 15k lines) interfaced with QepCAD and other SMT solvers, and validated it on more than 10 complex examples.

Computer Science Teacher
Sep 2015 to Sep 2017, part-time

Université Paris-Sud

During my Ph.D., taught 2nd- and 3rd-year computer science students. Courses included programming languages (C++), algorithms, software testing, networking and routing.

Research Intern
Mar 2015 to Aug 2015, end-of-master internship

French Commissariat of Atomic Energy (CEA)

Addressed the safety of a cyber-physical railway system by applying a contract-based design methodology and demonstrating it in a 3D graphical model. A Python tool checks whether a safety property is violated throughout an animated automatic-metro scenario.

Projects

Epilepsy in the brain

NYU Abu Dhabi with Cleveland Clinic Abu Dhabi

Collaboration between the NYUAD MEG lab and the epilepsy team at Cleveland Clinic Abu Dhabi (CCAD). Press: A Closer Look at Epilepsy in the Brain, NYU Abu Dhabi News, October 2026.

NeuroWaves
2024 to present

NYU Abu Dhabi

Documentation, experiment code and processing pipelines for the MEG and EEG labs: BIDS-compliant datasets with automated validation and continuous testing. neurowaves.readthedocs.io

Geomstats
2018 to 2022

with Stanford and Inria

Open-source Python package for geometry and statistics on manifolds. Third contributor, working on graph-structured data and learning methods for automata. geomstats.ai

Education

Doctor of Philosophy (Ph.D.), Computer Science
2015 to 2018

French Atomic Energy Commission & Université Paris-Saclay

Thesis: "Modeling and Qualitative Simulation of Hybrid Systems"

Master of Engineering, System Architecture & Embedded Systems
2014 to 2015

CentraleSupélec

"Design, Modeling and Architecture of Complex Industrial Systems"

Engineering Diploma, Electrical Engineering
2010 to 2015

Faculty of Engineering, Lebanese University

"Electrical Engineering and Industrial Computing"

Skills

Programming & Software Engineering	C, C++, Java, Python, PyTorch, NumPy, Linux, Travis CI, Git, PyCharm, Jenkins
Formal & Symbolic Methods, Solvers	SAT & SMT, UPPAAL, QepCAD, Z3, dReach
Neuroscience (MEG / EEG)	MNE-Python, FieldTrip, BESA, BIDS
System Architecture, Modeling & Simulation	MagicDraw (Cameo), Enterprise Architect, arKitect, Papyrus, Multisim, MATLAB & Simulink, LabVIEW (certified), Blender 3D
Web Development	HTML, CSS, Angular
Networking	Cisco CCNA 1, CCNA 2, CCNA 3

Languages

English (advanced), French (fluent), Arabic (fluent)

Publications

The full, current list is on Google Scholar.